

Optimal Prefix Ordering for Reverse-Hölder Cardinality Lower Bounds

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Abstract

The xBound framework of Stoian et al. [7] derives provable lower bounds on join cardinalities from a generalized reverse-Hölder inequality of Krasopoulos and Bougoffa [3]. The coefficient appearing in that inequality is *permutation-sensitive*: it is a product of factors built from prefix products of the ratios $r_i = M_i/m_i$ of the per-vector statistics, so the quality of the resulting bound depends on the order in which the vectors are supplied. The published formulation therefore proposes searching over factorially many orderings. We observe that this optimization has additional structure and is exactly solvable. Writing $f(x) = \sqrt{x} + 1/\sqrt{x}$ and $P_k(\pi)$ for the k -th prefix product of the permuted ratios, the permutation-dependent quantity is $D(\pi) = \prod_{k=2}^n f(P_k(\pi))$, and it occurs in the *denominator* of the resulting lower bound. We prove that D is minimized by sorting the ratios M_i/m_i in nondecreasing order, so the ordering search collapses to an $O(n \log n)$ sort followed by a single evaluation. We further characterize *all* optimal orderings at the level of values: an ordering is optimal precisely when its first two entries are the two smallest ratios in either order and all remaining entries are sorted; in particular every ordering is optimal when $n = 2$. We extend the result to certified statistics, showing that a direct outer enclosure $\tilde{r}_i = M_i^+/m_i^-$ both preserves the sorting theorem and yields a denominator no larger than the one obtained by bounding the two summands of each factor independently, so that for a common nonnegative numerator the resulting lower bound is at least as strong; a three-vector example shows that the independently propagated formulation is genuinely not solved by sorting a single scalar. The principal new optimization, equality, certified-enclosure, and counterexample results are formally verified in Lean 4/Mathlib. All statements concern the published mathematical formulation of the bound; we make no claim about the behavior of any unavailable implementation.

1 Introduction

Cardinality estimation is the component of a query optimizer most often blamed for catastrophic plan choices, and the failure mode that hurts most in production is *underestimation*: a plan built on the belief that an intermediate result is tiny is allocated too little memory and too little parallelism. Recent work by Stoian et al. [7] argues for attacking this asymmetry directly by computing provable *lower* bounds on join sizes, so that an optimizer can refuse to believe an underestimate. Their framework, xBound, obtains such bounds from reverse inner-product inequalities applied to prefixes of degree sequences, with a generalized reverse-Hölder inequality of Krasopoulos and Bougoffa [3] as its main analytic engine.

That inequality has a feature which is unusual among the classical inequalities used in this area: it is not symmetric in its inputs. Its coefficient is a product of factors $B_2, \dots, B_n$, where B_k

depends only on the statistics of the *first* k vectors. Consequently the bound one obtains depends on the order in which the n vectors are presented, and one is free to choose the order that gives the best bound. Stoian et al. [7] note this explicitly and propose evaluating the bound over all non-redundant permutations, keeping the maximum. This is a sound strategy, and it is the component we study here.

Our observation is that this particular ordering problem does not require search. After rearrangement the permutation-dependent quantity is a product of a fixed increasing function evaluated at the prefix products of the ratios $r_i = M_i/m_i \geq 1$, and prefix products of a sorted positive sequence are coordinatewise minimal among all orderings. The ordering optimization is therefore solved exactly by sorting. The mechanism is elementary, and in the logarithmic domain it is recognizably the same dominance argument that underlies shortest-processing-time scheduling [6]; we make no claim to have discovered it. What appears not to have been recorded is that the reverse-Hölder coefficient used by xBound has exactly this structure, together with the precise set of optimal orderings, the behavior under certified rather than exact statistics, and the boundary at which the sortable structure breaks down.

Contributions.

1. We identify a prefix-dominance structure in the generalized reverse-Hölder coefficient used by xBound and prove that sorting the ratios M_i/m_i in nondecreasing order minimizes the permutation-dependent denominator (Theorem 4.2).
2. Consequently the permutation optimization in the published formulation is solved exactly in $O(n \log n)$ time, replacing enumeration over orderings (Corollary 4.3).
3. We characterize all value-level optimal orderings, including ties, multiplicative coincidences, and the boundary case $n = 2$ (Theorem 4.5).
4. We examine the first-two-position symmetry described in the published formulation and observe that it generates $n!/2$ classes of labeled orderings (Proposition 4.8).
5. We extend the sorting theorem to certified outer statistics and prove that a direct outer enclosure yields a denominator no larger than independent propagation of the two summands, hence a lower bound at least as strong for a common nonnegative numerator (Theorems 5.4 and 5.5); a three-vector example shows the independently propagated formulation is not solved by sorting one scalar (Example 5.6).
6. We formally verify the principal new results in Lean 4/Mathlib (Section 6).

Statements marked $\checkmark$ are formally verified; Section 6 gives the theorem-to-declaration mapping and states precisely what is and is not covered by the artifact.

2 Background and notation

2.1 The generalized reverse-Hölder inequality

Throughout, $n \geq 2$ is the number of vectors and $d \geq 1$ their common length. We use i to index vectors, j to index coordinates, and k to index prefix lengths.

Theorem 2.1 (Krasopoulos–Bougoffa [3], Theorem 2.3, eq. (2.6); discrete form as used in [7]). *Let $v^{(1)}, \dots, v^{(n)} \in \mathbb{R}^d$ be positive vectors with*

$$0 < m_i \leq v_j^{(i)} \leq M_i < \infty, \quad i \in [n], j \in [d].$$

Then

$$\prod_{i=1}^n \|v^{(i)}\|_2 \leq \frac{(\sqrt{d})^{n-2}}{2^{n-1}} \left(\prod_{k=2}^n B_k \right) \left\| \prod_{i=1}^n v^{(i)} \right\|_1, \quad B_k = \sqrt{\prod_{i=1}^k \frac{M_i}{m_i}} + \sqrt{\prod_{i=1}^k \frac{m_i}{M_i}}. \quad (1)$$

We take Theorem 2.1 as a published external premise; it is not reproved here, and it is not part of our formal artifact. Two features of its statement matter for everything that follows. First, the factors B_k are defined only for $k \geq 2$, and the product in (1) likewise starts at $k = 2$; the length-one prefix never contributes. Second, B_k involves only the statistics of the first k vectors, which is precisely why the inequality is order-dependent. The original inequality of Krasopoulos and Bougoffa [3] is stated for functions on an interval $[a, b]$ with $(\sqrt{b-a})^{n-2}$ in place of $(\sqrt{d})^{n-2}$; its proof is an induction that combines the first $n-1$ functions and then adjoins the last using the Pólya–Szegő inequality [5], which is exactly what produces the asymmetric prefix chain. That paper contains no discussion of permuting the functions.

In xBound the vectors are prefixes of degree sequences of the joined columns, and d is the length of those prefixes, itself a certified lower bound on the number of distinct joining keys; we refer to Stoian et al. [7] for that reduction and treat (1) as given.

2.2 The permutation-dependent quantity

Since $d \geq 1$, the hypotheses of Theorem 2.1 force $m_i \leq v_j^{(i)} \leq M_i$ for at least one coordinate j , hence $0 < m_i \leq M_i$. We may therefore define

$$r_i = \frac{M_i}{m_i} \geq 1, \quad i \in [n]. \quad (2)$$

For a permutation π of $[n]$ — meaning that the vectors are supplied in the order $v^{(\pi(1))}, \dots, v^{(\pi(n))}$, so that the statistics are permuted along with them — put

$$P_k(\pi) = \prod_{j=1}^k r_{\pi(j)}, \quad f(x) = \sqrt{x} + \frac{1}{\sqrt{x}}, \quad D(\pi) = \prod_{k=2}^n f(P_k(\pi)). \quad (3)$$

Because all m_i, M_i are positive, $\sqrt{\prod_{i \leq k} m_{\pi(i)} / M_{\pi(i)}} = 1 / \sqrt{P_k(\pi)}$, so the k -th factor of (1) under the ordering π is exactly $B_k = f(P_k(\pi))$, and $D(\pi) = \prod_{k=2}^n B_k$.

Rearranging (1) to isolate the quantity that xBound wants to bound from below gives

$$\left\| \prod_{i=1}^n v^{(i)} \right\|_1 \geq \frac{2^{n-1}}{(\sqrt{d})^{n-2}} \cdot \frac{\prod_{i=1}^n \|v^{(i)}\|_2}{D(\pi)}. \quad (4)$$

Every quantity in (4) other than $D(\pi)$ is invariant under relabeling: the product $\prod_i \|v^{(i)}\|_2$ ranges over all i , the coordinatewise product $\prod_i v^{(i)}$ and hence its ℓ_1 norm is order-independent, and the prefactor involves only n and d . The same remains true if the exact norms $\|v^{(i)}\|_2$ are replaced by fixed lower estimates, as long as those estimates do not themselves depend on the ordering. We record the consequence explicitly, since the direction is easy to invert by accident.

Remark 2.2 (Optimization direction). $D(\pi)$ appears in the *denominator* of (4). Therefore *minimizing* $D(\pi)$ *maximizes* the resulting lower bound, and the best ordering is the one making D smallest.

3 Prefix-product dominance

The whole ordering result rests on the following elementary fact, which we state for a general positive sequence.

Lemma 3.1 (Prefix dominance[✓]). *Let $0 < x_1 \leq x_2 \leq \dots \leq x_n$ and let π be a permutation of $[n]$. Then*

$$\prod_{i=1}^k x_i \leq \prod_{j=1}^k x_{\pi(j)} \quad (0 \leq k \leq n).$$

Proof. For $k = 0$ both sides are empty products, equal to 1. Fix $1 \leq k \leq n$ and let $S = \{\pi(1), \dots, \pi(k)\}$, a k -element subset of $[n]$; write its elements in increasing order as $i_1 < i_2 < \dots < i_k$. Since $i_1, \dots, i_j$ are j distinct elements of $\{1, \dots, i_j\}$, we get the index certificate

$$i_j \geq j \quad (1 \leq j \leq k).$$

As x is nondecreasing, $x_{i_j} \geq x_j$ for every j , and all terms are positive, so multiplying these k inequalities gives $\prod_{j \leq k} x_j \leq \prod_{j \leq k} x_{i_j}$. Finally $\prod_{j \leq k} x_{\pi(j)} = \prod_{j \leq k} x_{i_j}$, since both are products over the same set S . $\square$

Only nonnegativity of the entries is used in the last step, so Lemma 3.1 holds verbatim for $0 \leq x_1 \leq \dots \leq x_n$; the formalization proves that weaker version and derives the positive case from it. Positivity becomes essential only when equality is analyzed, where one must cancel prefix products (Lemma 4.4).

The lemma says that the vector $(P_1(\text{id}), \dots, P_n(\text{id}))$ of sorted prefix products is *coordinatewise* minimal over all orderings, not merely minimal in some aggregate sense. This immediately yields a general optimization principle.

Theorem 3.2 (Monotone prefix objectives[✓]). *Let $0 < x_1 \leq \dots \leq x_n$, let π be a permutation of $[n]$, and let $g : (0, \infty) \rightarrow \mathbb{R}$ be nondecreasing. Then*

$$\sum_{k=2}^n g(P_k(\text{id})) \leq \sum_{k=2}^n g(P_k(\pi)),$$

and if additionally $g \geq 0$ on $(0, \infty)$,

$$\prod_{k=2}^n g(P_k(\text{id})) \leq \prod_{k=2}^n g(P_k(\pi)).$$

Proof. Every prefix product of a positive sequence is positive, so all arguments lie in the domain of g , and Lemma 3.1 gives $P_k(\text{id}) \leq P_k(\pi)$ for each k ; hence $g(P_k(\text{id})) \leq g(P_k(\pi))$ termwise. Summing preserves termwise inequalities with no further hypothesis. For the product, a termwise inequality between finitely many factors is preserved provided the smaller factors are nonnegative, which is the added hypothesis. $\square$

The nonnegativity hypothesis in the product case is not an artifact. Take two argument pairs $u = (1, 1)$ and $v = (2, 2)$, so that $u_k \leq v_k$ termwise, and let $g(z) = -1/z$, which is increasing on $(0, \infty)$ but negative there. Then

$$\prod_k g(u_k) = (-1)(-1) = 1 \quad \text{while} \quad \prod_k g(v_k) = \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right) = \frac{1}{4},$$

so the product inequality reverses. Sign changes destroy the product statement; the sum statement is unaffected.

Nothing in Theorem 3.2 is specific to the index range $2, \dots, n$: it holds for any subfamily of prefix lengths. We state it for $k \geq 2$ because that is the range occurring in (1), and the omission of $k = 1$ is exactly what produces the first-two-position freedom in Theorem 4.5.

4 Optimal ordering for the xBound coefficient

4.1 Monotonicity of the reverse-Hölder factor

Lemma 4.1 (Strict monotonicity of $f^\vee$). *$f(x) = \sqrt{x} + 1/\sqrt{x}$ is strictly increasing on $[1, \infty)$, and $f > 0$ there.*

Proof. Let $1 \leq x < y$. Then

$$f(y) - f(x) = (\sqrt{y} - \sqrt{x}) - \left(\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{y}} \right) = (\sqrt{y} - \sqrt{x}) - \frac{\sqrt{y} - \sqrt{x}}{\sqrt{xy}} = (\sqrt{y} - \sqrt{x}) \left(1 - \frac{1}{\sqrt{xy}} \right).$$

Since $y > x \geq 1$ we have $\sqrt{y} > \sqrt{x}$, and $xy > 1$ gives $\sqrt{xy} > 1$, so the second factor is positive. Hence $f(y) > f(x)$. Positivity is clear since both summands are positive for $x > 0$. $\square$

The factorization avoids any case analysis at the left endpoint. A derivative argument gives $f'(x) = (x - 1)/(2x^{3/2})$, which *vanishes* at $x = 1$, so establishing strictness at the endpoint from f' alone requires an extra step; the identity above does not.

4.2 The sorting theorem

Theorem 4.2 (Optimal ordering $^\vee$). *Let $1 \leq r_1 \leq r_2 \leq \dots \leq r_n$ and let π be any permutation of $[n]$. Then*

$$D(\text{id}) \leq D(\pi), \quad \text{where } D(\pi) = \prod_{k=2}^n f(P_k(\pi)).$$

Equivalently, sorting the ratios M_i/m_i in nondecreasing order minimizes the permutation-dependent coefficient of (1) and hence, by Remark 2.2, maximizes the lower bound (4).

Proof. Each $r_i \geq 1$, so every prefix product satisfies $P_k(\pi) \geq 1$ for every π and k ; in particular all arguments of f lie in $[1, \infty)$, where by Lemma 4.1 f is nondecreasing and positive. Lemma 3.1 gives $P_k(\text{id}) \leq P_k(\pi)$ for each k , whence $f(P_k(\text{id})) \leq f(P_k(\pi))$ factorwise. All factors on the smaller side are positive, so the product inequality follows by Theorem 3.2 applied with $g = f$ restricted to $[1, \infty)$. $\square$

Corollary 4.3 (Complexity). *Given m_i, M_i for $i \in [n]$, an ordering minimizing D is computed by forming the ratios $r_i = M_i/m_i$ in $O(n)$ time, sorting them in $O(n \log n)$ time, and evaluating the prefix chain $P_2, \dots, P_n$ once in $O(n)$ time. The total cost is $O(n \log n)$, and no enumeration over orderings is required.*

For numerically stable evaluation when the ratios are large, one may sort $\log r_i$, accumulate $s_k = \sum_{j \leq k} \log r_{\pi(j)}$, and use $\log f(e^{s_k}) = \frac{s_k}{2} + \log(1 + e^{-s_k})$.

We emphasize the scope of Theorem 4.2 and Corollary 4.3: they concern the ordering optimization induced by the published formulation (1)–(4). They say nothing about which orderings, if any, a particular implementation actually evaluates; see Section 8.

4.3 All optimal orderings

Theorem 4.2 exhibits an optimum. We now determine every optimum. Because equal ratios carried by different vectors are interchangeable in a way that is invisible to D , the right level for the characterization is the sequence of *values*. Let $x_1 \leq \dots \leq x_n$ denote the ratios in nondecreasing order, with multiplicity, and let $(y_1, \dots, y_n)$ be any rearrangement of this multiset; write $D(y) = \prod_{k=2}^n f(\prod_{j \leq k} y_j)$.

The following lemma is what rules out accidental numerical coincidences.

Lemma 4.4 (Equality in one prefix[✓]). *With x and y as above and all $x_i > 0$, and for any k ,*

$$\prod_{j=1}^k x_j = \prod_{j=1}^k y_j \quad \Longleftrightarrow \quad \{y_1, \dots, y_k\} = \{x_1, \dots, x_k\} \text{ as multisets.}$$

Proof. ($\Leftarrow$) Equal multisets have equal products. ($\Rightarrow$) Sort the selected values as $q_1 \leq \dots \leq q_k$. As in the proof of Lemma 3.1, $q_j \geq x_j$ for every j . All quantities are positive and the two products agree, so no inequality can be strict: if $q_{j_0} > x_{j_0}$ for some j_0 , then multiplying $q_j \geq x_j > 0$ over all j with strictness at j_0 would give $\prod_j q_j > \prod_j x_j$, a contradiction. Hence $q_j = x_j$ for all j , which is the claimed multiset equality. $\square$

Theorem 4.5 (Exact minimizers[✓]). *Let $1 \leq x_1 \leq \dots \leq x_n$ with $n \geq 2$, and let $(y_1, \dots, y_n)$ be a rearrangement of the multiset $\{x_1, \dots, x_n\}$. Then*

$$D(y) = D(x) \quad \Longleftrightarrow \quad \{y_1, y_2\} = \{x_1, x_2\} \text{ as multisets} \quad \text{and} \quad y_k = x_k \quad (3 \leq k \leq n).$$

Combined with Theorem 4.2, these are exactly the minimizing value sequences.

Proof. ($\Leftarrow$) Assume the stated normal form. For every $k \geq 2$ the multiset $\{y_1, \dots, y_k\}$ equals $\{x_1, \dots, x_k\}$: the first two entries agree as a multiset by hypothesis, and the entries in positions $3, \dots, k$ agree entrywise. Hence $\prod_{j \leq k} y_j = \prod_{j \leq k} x_j$ for all $k \geq 2$, so every factor of D agrees and $D(y) = D(x)$. Note that position 1 alone is never constrained, because the product defining D starts at $k = 2$.

($\Rightarrow$) By Lemma 3.1 and Lemma 4.1, for every k with $2 \leq k \leq n$,

$$0 < f\left(\prod_{j \leq k} x_j\right) \leq f\left(\prod_{j \leq k} y_j\right),$$

all prefix products lying in $[1, \infty)$ because every $x_i \geq 1$. A product of finitely many positive factors, each at most the corresponding factor on the other side, is strictly smaller as soon as one of the inequalities is strict. Since $D(x) = D(y)$, no inequality is strict, so

$$f\left(\prod_{j \leq k} x_j\right) = f\left(\prod_{j \leq k} y_j\right) \quad (2 \leq k \leq n).$$

By Lemma 4.1, f is injective on $[1, \infty)$, so $\prod_{j \leq k} x_j = \prod_{j \leq k} y_j$ for all $2 \leq k \leq n$. Applying Lemma 4.4 at $k = 2$ gives $\{y_1, y_2\} = \{x_1, x_2\}$. For $3 \leq k \leq n$, the equalities at $k-1$ and k give $(\prod_{j \leq k-1} x_j)x_k = (\prod_{j \leq k-1} y_j)y_k$ with $\prod_{j \leq k-1} x_j = \prod_{j \leq k-1} y_j > 0$; cancelling this positive factor yields $y_k = x_k$. $\square$

Remark 4.6 (Reading the characterization). Four consequences are worth stating explicitly.

1. *The first two positions may be exchanged freely.* The product defining D omits $k = 1$, so P_1 never enters; every factor that does enter contains both y_1 and y_2 . This is the same symmetry noted by Stoian et al. [7].
2. *Ties are handled automatically.* The characterization constrains values, not labels, so vectors carrying equal ratios may be permuted arbitrarily among the positions holding that value.
3. *For $n = 2$ every ordering is optimal.* The only factor is $k = 2$, whose argument $r_1 r_2$ is the full product, hence order-independent; correspondingly the tail condition of Theorem 4.5 is vacuous.
4. *Multiplicative coincidences create no extra minimizers.* One might worry that distinct value multisets with equal products (say $\{1, 6\}$ and $\{2, 3\}$) could give a second optimum. Lemma 4.4 excludes this: equality of a positive prefix product forces multiset equality, not merely numerical equality.

Remark 4.7 (Counting labeled minimizers; not formalized). If the distinct ratio values have multiplicities m_v , the number of *labeled* permutations attaining the optimum is $2 \prod_v m_v!$ when $x_1 < x_2$, and $\prod_v m_v!$ when $x_1 = x_2$: there are two normal-form value sequences in the first case and one in the second, and each admits $\prod_v m_v!$ labelings. This bookkeeping consequence is elementary but is not part of the formal artifact.

4.4 The first-two-position symmetry and its class count

Stoian et al. [7, Sec. 2.2.1] describe the search space by excluding permutations that differ only by swapping the first two elements, and state that $(n - 1)!$ permutations remain. The following elementary count records what that symmetry alone yields.

Proposition 4.8 (Class count; not formalized). *Let τ act on the set of orderings by exchanging the entries in positions 1 and 2. Then τ is an involution without fixed points on the $n!$ labeled orderings, so its orbits all have size exactly 2 and the number of equivalence classes is $n!/2$. For $n = 2$ this equals $(n - 1)! = 1$; for every $n > 2$ it is strictly larger, since $n!/2 = (n - 1)! \cdot n/2 > (n - 1)!$.*

Proof. An ordering assigns distinct labels to positions 1 and 2, so exchanging them always produces a different ordering; hence τ has no fixed point and every orbit has size 2. The orbit count is therefore $n!/2$. The comparison with $(n - 1)!$ is immediate. $\square$

Concretely, for three vectors with distinct ratios $a < b < c$ the objective is determined by which pair occupies the first two positions, since the $k = 3$ factor $f(abc)$ is common to all orderings; the three attainable values $f(ab)f(abc)$, $f(ac)f(abc)$, $f(bc)f(abc)$ are pairwise distinct by Lemma 4.1, giving three classes rather than $(3 - 1)! = 2$. A count of $(n - 1)!$ does arise if one *fixes* a designated vector in the first position, but that restriction is not without loss of generality: for ratios 2, 3, 5, fixing the vector with ratio 5 first makes the optimal first pair $\{2, 3\}$ unreachable.

We state this narrowly. The first-two-position symmetry described in the published account yields $n!/2$ equivalence classes rather than $(n - 1)!$; this concerns the stated combinatorial justification and implies nothing about the unavailable implementation. It is also worth noting that Theorem 4.2 makes the exact size of the search space moot, since no enumeration is needed in the first place.

5 Certified statistics

In practice the exact extrema $m_i^* = \min_j v_j^{(i)}$ and $M_i^* = \max_j v_j^{(i)}$ may be unavailable, and a system instead holds certified bounds on them. This section shows the sorting theorem survives the natural certified formulation, and compares that formulation with a plausible alternative.

5.1 Direct outer enclosure

Definition 5.1 (Direct outer certificate). A *direct outer certificate* for $v^{(i)}$ is a pair (m_i^-, M_i^+) with

$$0 < m_i^- \leq v_j^{(i)} \leq M_i^+ < \infty \quad \text{for every coordinate } j.$$

Theorem 2.1 requires valid positive lower and finite upper bounds on the coordinates; it does not require them to be attained. It may therefore be applied verbatim with $\tilde{m}_i = m_i^-$ and $\tilde{M}_i = M_i^+$.

Proposition 5.2 (Certified ratios[✓]). For a direct outer certificate, $\tilde{r}_i = M_i^+/m_i^- \geq 1$.

Proof. Since $d \geq 1$ there is a coordinate j , and $m_i^- \leq v_j^{(i)} \leq M_i^+$ with $m_i^- > 0$ gives $M_i^+/m_i^- \geq 1$. $\square$

Consequently Theorem 4.2 applies unchanged to the ratios $\tilde{r}_i$: sorting M_i^+/m_i^- nondecreasingly and evaluating once gives, in $O(n \log n)$ time, the best bound obtainable from this direct-enclosure formulation. If $m_i^- = 0$ the construction is unavailable, as the coefficient is then not finite; a separate argument, such as positivity of integral degrees, may justify replacing a zero lower bound by 1.

The price of using an enclosure rather than exact extrema is a weaker but still valid coefficient, quantified as follows. Fix a prefix and write

$$R = \prod_i \frac{M_i^*}{m_i^*}, \quad A = \prod_i \frac{M_i^+}{m_i^-}.$$

Theorem 5.3 (True versus outer factor[✓]). Suppose $0 < m_i^- \leq m_i^* \leq M_i^* \leq M_i^+$ for every i in the prefix. Then $1 \leq R \leq A$, and consequently $f(R) \leq f(A)$.

Proof. Factorwise, $m_i^* \leq M_i^*$ with $m_i^* \geq m_i^- > 0$ gives $M_i^*/m_i^* \geq 1$, so $R \geq 1$ as a product of numbers at least 1. Also $M_i^* \leq M_i^+$ and $0 < m_i^- \leq m_i^*$ give $M_i^*/m_i^* \leq M_i^+/m_i^-$ factorwise, and all factors are positive, so $R \leq A$. Both lie in $[1, \infty)$, so Lemma 4.1 yields $f(R) \leq f(A)$. $\square$

5.2 Independent propagation of the two summands

A different way to use interval statistics is to bound the two summands of B_k separately. Suppose the system stores four endpoints per vector,

$$0 < m_i^- \leq m_i^* \leq m_i^+ < \infty, \quad 0 < M_i^- \leq M_i^* \leq M_i^+ < \infty,$$

and, for a fixed prefix, sets

$$A = \prod_i \frac{M_i^+}{m_i^-}, \quad C = \prod_i \frac{m_i^+}{M_i^-}, \quad B_{\text{ind}} = \sqrt{A} + \sqrt{C}.$$

Since $\sqrt{R} \leq \sqrt{A}$ and $1/\sqrt{R} \leq \sqrt{C}$, this is a valid upper bound on $B^* = f(R)$. It is, however, a *two-parameter* quantity: A and C are in general not reciprocal, and each depends on the prefix. Write $B_{\text{enc}} = f(A) = \sqrt{A} + 1/\sqrt{A}$ for the direct-enclosure factor.

Theorem 5.4 (Direct enclosure dominates independent propagation[✓]). *With the four-endpoint certificates above,*

$$AC = \prod_i \left(\frac{M_i^+}{M_i^-} \right) \left(\frac{m_i^+}{m_i^-} \right) \geq 1, \quad \text{hence} \quad \frac{1}{\sqrt{A}} \leq \sqrt{C},$$

and therefore

$$B_{\text{enc}} = \sqrt{A} + \frac{1}{\sqrt{A}} \leq \sqrt{A} + \sqrt{C} = B_{\text{ind}}.$$

Proof. Regrouping the two products factorwise gives the displayed formula for AC . Each factor $M_i^+/M_i^- \geq 1$ because $0 < M_i^- \leq M_i^+$, and each factor $m_i^+/m_i^- \geq 1$ because $0 < m_i^- \leq m_i^+$; a product of numbers at least 1 is at least 1, so $AC \geq 1$. Since $A > 0$ and $C \geq 0$, the inequality $AC \geq 1$ gives $\sqrt{AC} \geq 1$, i.e. $\sqrt{A}\sqrt{C} \geq 1$, i.e. $1/\sqrt{A} \leq \sqrt{C}$. Adding $\sqrt{A}$ to both sides gives the conclusion. $\square$

Note the asymmetry of hypotheses: the direct enclosure needs only $m_i^- > 0$, whereas forming C additionally requires $M_i^- > 0$. The comparison is factorwise and holds for every prefix and every ordering, so it lifts to whole denominators. Writing $D_{\text{enc}}(\pi) = \prod_{k=2}^n f(A_k(\pi))$ and $D_{\text{ind}}(\pi) = \prod_{k=2}^n (\sqrt{A_k(\pi)} + \sqrt{C_k(\pi)})$, where A_k, C_k are the corresponding k -prefix products, we obtain $D_{\text{enc}}(\pi) \leq D_{\text{ind}}(\pi)$ for every π , all factors being positive.

Theorem 5.5 (Same-numerator lower-bound comparison[✓]). *Let $N \geq 0$ be a numerator estimate that does not depend on the ordering, and let π_s sort the scalars $\tilde{r}_i = M_i^+/m_i^-$ nondecreasingly. Then for every permutation π ,*

$$\frac{N}{D_{\text{ind}}(\pi)} \leq \frac{N}{D_{\text{enc}}(\pi)} \leq \frac{N}{D_{\text{enc}}(\pi_s)}.$$

In particular $N/D_{\text{enc}}(\pi_s) \geq \max_{\pi} N/D_{\text{ind}}(\pi)$.

Proof. All denominators are positive, being finite products of positive factors. The first inequality is $D_{\text{enc}}(\pi) \leq D_{\text{ind}}(\pi)$ together with antitonicity of $t \mapsto N/t$ on $(0, \infty)$ for $N \geq 0$. The second is Theorem 4.2 applied to the ratios $\tilde{r}_i$, which are at least 1 by Proposition 5.2, again with antitonicity. $\square$

The hypothesis that N is the *same* on both sides is essential and is stated as such: the theorem compares two denominators against one common numerator. If two formulations use different numerator estimates, comparing denominators alone says nothing about the resulting lower bounds.

5.3 Sorting one scalar fails for independent propagation

Theorem 4.2 applies to D_{enc} because that objective depends on the ordering through the single scalar sequence $\tilde{r}_i$. The independently propagated objective D_{ind} depends on two accumulated products, and the adjacent-exchange argument no longer transfers. The following exact example shows the failure is real, already for the smallest interesting size: $n = 2$ is order-invariant, so $n = 3$ is minimal.

Example 5.6 (Independent propagation is not sorted by $\tilde{r}^\vee$). Take three vectors of length $d = 2$,

$$X = (1, 1), \quad U = (1, 1), \quad V = (4, 25),$$

with certificates as follows. For X and V all four endpoints are exact. For U the minimum endpoints are exact, $m_U^- = m_U^+ = 1$, while the maximum is known only through the valid but loose certificate $1 = M_U^- \leq M_U^* \leq M_U^+ = 4$. All certificates are valid. The resulting pairs $(\tilde{r}_i, c_i) = (M_i^+/m_i^-, m_i^+/M_i^-)$ are

$$X : (1, 1), \quad U : (4, 1), \quad V : \left(\frac{25}{4}, \frac{4}{25}\right),$$

so the scalars $\tilde{r}_i$ are strictly increasing in the order X, U, V . Because $n = 3$, the $k = 3$ factor is common to all orderings; it equals $\sqrt{25} + \sqrt{4/25} = 5 + \frac{2}{5} = \frac{27}{5}$. The $k = 2$ factors are

$$B_{XU} = \sqrt{4} + \sqrt{1} = 3, \quad B_{XV} = \sqrt{\frac{25}{4}} + \sqrt{\frac{4}{25}} = \frac{5}{2} + \frac{2}{5} = \frac{29}{10}, \quad B_{UV} = \sqrt{25} + \sqrt{\frac{4}{25}} = \frac{27}{5}.$$

Hence the complete objectives are

$$D_{\text{ind}}(X, U, V) = 3 \cdot \frac{27}{5} = \frac{81}{5} = 16.2, \quad D_{\text{ind}}(X, V, U) = \frac{29}{10} \cdot \frac{27}{5} = \frac{783}{50} = 15.66.$$

Sorting by $\tilde{r}_i$ selects the ordering X, U, V , which is strictly worse than X, V, U . So no single-scalar sort optimizes D_{ind} .

Two comments place this example correctly. First, it does not contradict Theorem 4.2: that theorem is about D_{enc} , and indeed for these same certificates the sorted direct enclosure gives $D_{\text{enc}}(\pi_s) = f(4)f(25) = \frac{5}{2} \cdot \frac{26}{5} = 13$, which is smaller than *every* value of D_{ind} , as Theorem 5.5 predicts. Second, the example marks a genuine boundary: the two-parameter ordering problem for D_{ind} is a different optimization problem, and we do not treat it here. Given Theorem 5.5, the practical reading is that the direct-enclosure formulation is the preferable one whenever both are available, since it is simultaneously sortable and at least as tight.

6 Formal verification

The results marked $\checkmark$ above are formalized in Lean 4 [1] on top of Mathlib [8]. The artifact accompanies this paper.

Environment and checks. The development builds against Lean 4.32.2 and Mathlib at tag v4.32.2, commit 905b95818eb32af7874a58b427f50c1711a5e96c. Both

```
lake build    and    lake env lean Main.lean
```

complete successfully. A scan of the project sources, excluding the `.lake` dependency directory, finds no occurrence of `sorry`, `admit`, `unsafe`, `native_decide`, or any `axiom` declaration. Running `#print axioms` on each principal result reports only the three standard axioms of Lean’s logic, `propext`, `Classical.choice`, and `Quot.sound`; the development introduces no axiom of its own.

What is and is not formalized. Theorem 2.1, the generalized reverse-Hölder inequality itself, is taken as a published external premise and is *not* formalized: the artifact verifies the ordering, equality, certified-enclosure, and counterexample results that follow once that inequality is assumed, not the underlying analytic inequality. Two secondary items are also deliberately informal: the class count of Proposition 4.8 and the labeled-minimizer count of Remark 4.7. Accordingly we claim only that the principal new optimization, equality, certified-enclosure, and counterexample results are formally verified, not that every statement in this paper is.

Result	Lean declaration	Module
Lemma 3.1 (tuple)	<code>sortedPrefixProduct_le_prefixProduct</code>	<code>PrefixProducts</code>
Lemma 3.1 (list)	<code>sorted_take_prod_le_of_perm</code>	<code>PrefixProducts</code>
Lemma 4.4	<code>sorted_take_prod_eq_iff_perm_take</code>	<code>PrefixProducts</code>
Theorem 3.2 (product)	<code>sorted_prefixProductObjective_le</code>	<code>MonotoneObjective</code>
Theorem 3.2 (sum)	<code>sorted_prefixSumObjective_le</code>	<code>MonotoneObjective</code>
Lemma 4.1	<code>reverseHolderFactor_strictMonoOn</code>	<code>XBoundFactor</code>
Theorem 4.2	<code>sorted_xBoundObjective_le</code>	<code>Ordering</code>
Theorem 4.2 (list)	<code>sorted_listXBoundObjective_le</code>	<code>Ordering</code>
Theorem 4.5	<code>listXBoundObjective_eq_iff_valueNormalForm</code>	<code>Equality</code>
Remark 4.6(3), $n = 2$	<code>listXBoundObjective_eq_of_length_two</code>	<code>Equality</code>
Proposition 5.2	<code>one_le_ratio_of_direct_outer_certificate</code>	<code>CertifiedIntervals</code>
Thm 5.3, $1 \leq R \leq A$	<code>one_le_true_product_and_le_outer</code>	<code>CertifiedIntervals</code>
Thm 5.3, $f(R) \leq f(A)$	<code>certified_true_factor_le_outer</code>	<code>CertifiedIntervals</code>
Theorem 5.4	<code>certified_enclosure_factor_le_independent</code>	<code>CertifiedIntervals</code>
Denominator dominance	<code>sorted_xBoundObjective_le_independentIntervalObjective</code>	<code>CertifiedIntervals</code>
Theorem 5.5	<code>independent_lowerBound_le_sorted_direct</code>	<code>CertifiedIntervals</code>
Example 5.6	<code>naive_interval_full_objective_counterexample</code>	<code>Counterexample</code>

Table 1: Correspondence between the results marked ✓ and the Lean declarations. All declarations are in the `XBound` namespace and are imported by `XBound.lean`.

Mapping. Table 1 lists the correspondence for the results marked ✓. All declarations live in the `XBound` namespace. Two formulations are maintained in parallel: a finite-tuple form using a monotone $x : \text{Fin } n \rightarrow \mathbb{R}$ with a permutation `Equiv.Perm (Fin n)`, and a list form using `xs.Pairwise` for the order relation together with `ys.Perm xs`. Appendix A lists the supporting declarations.

Two points about the formalization deserve mention. The counterexample of Example 5.6 is verified at the level of the *complete* three-vector objective, including the order-invariant $k = 3$ factor, rather than only for the first-pair comparison; the exact rational values 3, 29/10, and 27/5 are separately checked. Theorem 5.5 uses literally the same Lean variable `N` on both sides, so it cannot be misread as comparing two different numerator estimates, and it assumes only $0 \leq N$.

Reproducing the build. In a clone of the artifact repository, with `elan` available, the toolchain is pinned by the `lean-toolchain` file and the Mathlib dependency by `lakefile.toml`. The build steps are:

```
lake exe cache get      # optional: prebuilt Mathlib oleans
lake build
lake env lean Main.lean
```

We ran this sequence in a fresh clone: dependency resolution from the tag `v4.32.2` produced exactly the commit recorded above, and all 1515 build jobs completed from clean sources, after which `Main.lean` elaborated successfully. Because `lakefile.toml` pins Mathlib by tag rather than by commit, we recommend committing the generated `lake-manifest.json` alongside the sources, so that the exact commit, and not merely the tag, is fixed for future clones.

7 Related work

The inequality and its source. The analytic input is Theorem 2.3, equation (2.6), of Krasopoulos and Bougoffa [3], proved by induction: the first $n - 1$ functions are combined and the last is

adjoined via the Pólya–Szegő inequality [5], together with the norm comparison $\|h\|_1 \leq \sqrt{b-a} \|h\|_2$. That induction is precisely what makes the coefficient a prefix chain and hence order-dependent. The paper states the inequality for n functions and compares it against other reverse Hölder and Minkowski inequalities; it contains no permutation optimization or ordering discussion.

xBound. Stoian et al. [7] apply the discrete form of this inequality to prefixes of degree sequences to obtain provable join-size lower bounds, observe that the inequality is permutation-sensitive, and propose maximizing over non-redundant orderings. Our Theorem 4.2 concerns exactly that maximization. The other estimators they describe, including the min-degree bound, are symmetric in the relation labels, and the displayed reverse-Hölder term is the order-dependent one; their optional subset-dynamic-programming bound is a different construction that our result does not address.

Classical ordering results. Under the substitution $p_i = \log r_i \geq 0$, the logarithms of the prefix products $\log P_k(\pi) = \sum_{j \leq k} p_{\pi(j)}$ are the completion times of a single machine processing jobs in the order π . Ascending r_i is then the shortest-processing-time order, whose completion-time vector is coordinatewise minimal; the adjacent-interchange technique for such problems goes back to Smith [6], whose rule addresses the weighted total completion time. Our objective is different — a product of a fixed nonlinear function of the completion times, with the first term omitted — and we do not claim that Smith’s rule or the scheduling literature contains Theorem 4.2 or Theorem 4.5; the point is that the underlying dominance mechanism (Lemma 3.1) is classical in spirit. The same fact can be phrased in the vocabulary of majorization [4], where coordinatewise domination of prefix products is a log-majorization statement, though the standard rearrangement and majorization theorems are a less direct fit than the prefix/completion-time formulation. Within databases, Hellerstein and Stonebraker [2] use a Smith-style adjacent interchange to order expensive predicates under a multiplicative-prefix cost; the objective and the ranking key there differ from ours, but it is a precedent for this style of argument in query optimization.

Novelty. We searched for prior statements of the specific results above and did not locate one, but a negative search cannot establish novelty, and citation databases may lag for a recent preprint. We therefore classify the contribution conservatively: the ordering principle is elementary and classical in spirit, and what we claim is its application to the reverse-Hölder coefficient used by xBound, together with the complete set of optimal orderings, the certified-statistics comparison, the delimitation of where the sortable structure ends, and the formal verification of these.

8 Discussion and limitations

Scope of the algorithmic claim. Theorem 4.2 and Corollary 4.3 apply to the ordering optimization induced by the published formulation (1)–(4). They imply that this component need not be searched. They do not describe the behavior of any deployed system.

The implementation is not publicly inspectable. We were able to audit the mathematics because the paper and its analytic source are public. We were not able to audit an implementation. The public xBound demonstration repository contains an Electron demonstration together with precomputed benchmark results; its custom-query script is a wrapper that imports the estimator from an external source tree which is not present in the public repositories. Accordingly we make no claim about whether any implementation enumerates factorially many orderings, whether it uses

direct outer enclosures or independent interval propagation, or whether it would exhibit a measured speedup from Theorem 4.2. Each such claim would require source access. What can be said from public evidence is that the published formula is permutation-sensitive, that the published account describes a search over orderings, and that the displayed objective is exactly solvable by sorting.

Where the result does not reach. The theorem changes only the ordering loop for the displayed reverse-Hölder component. It does not affect the construction of statistics or sketches, norm retrieval, the min-degree bound, the heavy-partition machinery, the final maximum over estimators, or the optional subset-dynamic-programming bound. It also assumes $r_i \geq 1$, which holds automatically for ratios of valid bounds; without a condition of this kind the sorted order can fail, since f decreases on $(0, 1)$. Finally, Example 5.6 shows that the two-parameter objective arising from independent interval propagation is a genuinely different optimization problem, which we leave to future work; Theorem 5.5 suggests that moving to the direct-enclosure formulation is preferable to solving it.

9 Conclusion

The permutation-sensitivity of the generalized reverse-Hölder coefficient used by xBound is a real feature of the inequality, and choosing a good ordering genuinely improves the bound. We have shown that the resulting optimization problem is not a search problem: the coefficient is a product of an increasing function evaluated at prefix products of the ratios M_i/m_i , sorted prefix products are coordinatewise minimal, and therefore sorting the ratios in nondecreasing order is optimal, in $O(n \log n)$ time. The optimal orderings are exactly those whose first two entries are the two smallest ratios in either order with the remainder sorted, ties included and multiplicative coincidences excluded. The same statement survives replacing exact extrema by direct positive outer certificates, and that formulation dominates independent propagation of the two summands under a common numerator, while a three-vector example shows independent propagation is not itself solved by sorting a single scalar. Apart from the complexity statement and the two counting remarks, these results are formally verified in Lean 4/Mathlib. We hope the observation is useful to the xBound framework, whose overall approach to provable lower bounds motivated this note.

Acknowledgments. The formalization uses Lean 4 and Mathlib.

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A Supporting Lean declarations

Table 1 lists the declarations corresponding to the numbered results. The development contains the following further declarations, in the `xBound` namespace, which carry the intermediate steps.

Declaration	Role
<code>finIndex_le_of_strictMono</code>	the index certificate $i_j \geq j$ of Lemma 3.1
<code>sortedPrefixProduct_le_finset_of_nonneg</code>	prefix dominance against an arbitrary k -subset
<code>sortedPrefixProduct_le_selection_of_nonneg</code>	prefix dominance against an injective selection
<code>sortedPrefixProduct_le_prefixProduct_of_nonneg</code>	nonnegative form of Lemma 3.1
<code>prefixProductAt_refl_le</code>	coordinatewise sorted-prefix domination
<code>sorted_take_prod_le_of_perm_nonneg</code>	nonnegative list form
<code>forall2_prod_le_prod_of_nonneg</code>	coordinatewise comparison implies product order
<code>forall2_eq_of_prod_eq_of_pos</code>	positive equal products force coordinatewise equality
<code>finset_prod_monotoneOn_objective</code>	product form of Theorem 3.2
<code>finset_sum_monotoneOn_objective</code>	sum form of Theorem 3.2
<code>reverseHolderFactor_monotoneOn</code>	weak monotonicity used by the ordering corollary
<code>reverseHolderFactor_pos</code>	positivity of f on positive arguments
<code>one_le_prefixProductAt</code>	prefix products are at least 1 when $r_i \geq 1$
<code>one_le_certified_outer_ratio</code>	$M^+/m^- \geq 1$ from a four-endpoint certificate
<code>true_ratio_le_outer_ratio</code>	scalar form of Theorem 5.3
<code>one_div_sqrt_le_sqrt</code>	$1/\sqrt{A} \leq \sqrt{C}$ from $AC \geq 1$
<code>enclosure_factor_le_independent</code>	scalar form of Theorem 5.4
<code>one_le_certified_interval_products</code>	$AC \geq 1$ on a finite prefix
<code>xBoundObjective_le_independentIntervalObjective</code>	factorwise denominator dominance
<code>commonNumerator_div_antitone</code>	antitonicity of $t \mapsto N/t$ for $N \geq 0$
<code>independentPairFactor_XU/XV/UV</code>	the exact values 3, 29/10, 27/5
<code>naive_interval_sorting_counterexample</code>	the first-pair comparison of Example 5.6

Table 2: Supporting declarations in the formal development.

B Definitions used in the formalization

For reference, the central definitions are as follows, with $x : \text{Fin } n \rightarrow \mathbb{R}$ the ratio tuple and π a permutation.

$$\begin{aligned}
\text{prefixProduct } x \ \pi \ k &= \prod_{i < k} x(\pi(i)), \\
\text{sortedPrefixProduct } x \ k &= \prod_{i < k} x(i), \\
\text{objectiveIndices } n &= \{k \in \text{Fin}(n+1) : 2 \leq k\}, \\
\text{reverseHolderFactor } x &= \sqrt{x} + 1/\sqrt{x}, \\
\text{prefixProductObjective } g \ x \ \pi &= \prod_{k \in \text{objectiveIndices}} g(P_k), \\
\text{xBoundObjective } x \ \pi &= \prod_{k \in \text{objectiveIndices}} f(P_k), \\
\text{listXBoundObjective } xs &= \prod_{k=2}^{|xs|} f(\prod(xs.\text{take } k)), \\
\text{valueNormalForm } xs \ ys &= (ys.\text{take } 2 \sim xs.\text{take } 2) \ \wedge \ (ys.\text{drop } 2 = xs.\text{drop } 2).
\end{aligned}$$

The index set `objectiveIndices` implements the range $k = 2, \dots, n$ of (1); for $n \leq 1$ it is empty and the objectives are empty products equal to 1. The predicate `valueNormalForm` is exactly the right-hand side of Theorem 4.5: a multiset equality on the first two entries and an entrywise equality on the tail.